Chunyang Luo

Email: 2300011193@stu.pku.edu.cn Phone: 19990529007 GitHub: https://1lcy007.github.io/

Research Interests: Agent Systems | Multi-Robot Planning | LLM-based Planning & Reasoning | Embodied AI

EDUCATION

B.Eng. Candidate in Robotics Engineering, Peking University 2023 – Present

Expected Graduation: June 2027; GPA: 3.526/4.0 (85/100); Rank: 5/19 (Top 26%)

Selected Core Courses: Foundations of Machine Learning (98), Probability and Statistics (94), Introduction to Robotics (89)

RESEARCH EXPERIENCES

Multi-Robot Collaborative LTL Continuous Planning for Dynamic Environments Jul. 2025 – Present

Supervisor: Prof. Zhongkui Li

- Designed an online task decomposition and replanning framework for heterogeneous multi-robot LTL tasks under dynamic events.
- Built a heterogeneous multi-robot simulation platform (Unitree Go2, Scout) with task execution and state feedback interfaces.
- Benchmarked against static and periodic-replanning baselines on completion rate, planning time, and cumulative cost; currently extending to physical robots.

OpenGRAIL: Integrating LLM and Classical Planning for Multi-Robot Task Planning in Open Worlds Sept. 2025 – Present

Supervisor: Prof. Zhongkui Li

- Developed an LLM–classical planning framework that translates natural-language instructions into executable multi-robot plans.
- Designed hierarchical task decomposition and MILP-based robot allocation/scheduling under capability, temporal, spatial, and collaboration constraints.
- Integrated classical planning with AI2-THOR for closed-loop execution and environment feedback. **Manuscript in preparation for ICRA 2027.**

RepoFlow: A Repository-Level Workflow Framework for AI Coding Agents Mar. 2026 – Present

Supervisor: Prof. Kun Yuan

- Investigated context loss and repository degradation in long-horizon AI coding agents.
- Designed RepoFlow, a repository-level workflow framework with persistent context and structured Agent Handoff for cross-agent task continuity.
- Developed style-learning and code-health-review pipelines for persistent constraints and feedback across agent iterations.

AWARDS & COMPETITIONS

- **Second Prize**, 22nd “Jiang Zehan Cup” Modeling and Computer Applications Competition, Peking University
- **Honorable Mention**, MCM/ICM 2026

PROFESSIONAL TRAINING & INTERNSHIP

Alibaba DAMO Academy – Embodied Intelligence Bootcamp Spring 2026

Built a visual-navigation demo on a robotic arm, integrating perception, control, and task execution.

Xbotics Online Internship Spring 2026

Implemented RL-based robot navigation and explored Sim-to-Real transfer.

SKILLS AND OTHERS

- **Programming & Tools:** Python, MATLAB, Linux, Git, LaTeX
- **Planning & AI:** LTL, PDDL, Classical Planning, Multi-Robot Task Allocation, LLM-based Planning & Reasoning, RL
- **Robotics:** ROS, Gazebo, RViz, AI2-THOR
- **Language:** CET-6 578, IELTS Overall 7.0